The Digi-Comp II: what the simulation showed

West bay 3: the Digi-Comp II · 27 October 2026

1/2 inch steel balls (12.7 mm, 8.35 g), PLA parts, friction 0.3, no bounce, the board at 15 degrees, 2400 steps a second, the pins' friction 0.4 kg cm^2/s^2, unless a line says otherwise. Each mechanism was worked out alone first, in `tests/`, then on the board, with its tour.

Step 2b: adding and multiplying

Laying it out

Thirteen flip-flops, seven switches and the lanes between them have to fit on one level, and lanes cannot cross there (a ball can go down to the lower level, never back up). The board was laid out in a plan drawn from the geometry (every wall, hole and peg, the lanes coloured by kind) and changed until nothing touched. These settled most of it:

- **The order of the memory.** The memory's switches stand beside A1 to A4, down the accumulator's staircase, so the lanes reaching them, from above and the left, must keep their order: the leftmost of the distributor's four outlets can only go to the lowest, M4. So D2's left goes to the MQ and on to M4, D2's right to M3, D3's left to M2, D3's right to M1, and with the distributor set to start, the balls of a cycle go to the MQ (and M4), M1, M3, M2. The manual's order after the MQ is M2, M3, M1; the sum is the same.
- **The distributor is a tree,** not the original's ring: D1 sends the balls to D2 and D3 by turns, each of them by turns to its two sides. D2 is where the next step of a staircase would be, so D1's left drops straight into its funnel, and D2's left straight into MQ1's; D1's right, and every other outlet, falls into a **catch** (a funnel where the next flip-flop's would be) and runs on down a lane. D3 must stand far enough from D2 that D2's right catch and D3's left don't overlap (8 cm right of D1), and low enough that the lane to it falls steeply enough (14.2 cm below D1).
- **Lanes** fall at least 35 degrees from across the board (the geometry warns of any that don't); the flattest, D1's to D3, falls 39 degrees. The longest are MULTIPLY's down to CLEAR and D2's to M3, 27 cm each.

The board grew from 322 x 878 to 424 x 1216 mm (14 tiles a level), and its table from 100 x 150 to 110 x 180 cm.

Two ways into one funnel

A1 to A4 each take balls from two places: the carry from the flip-flop above (or COUNT, for A1) and the memory. The carry was traced first (a copy of `tests/toggle.lua` printing the ball's centre every 1/60 s): it leaves the cup moving 11 cm/s across and crosses the next funnel's mouth 4 mm left of its centre, so its left half is not free for a lane ending over it. Instead each of these funnels has a **long left wall**, out 35 mm and up 15.7 mm (31 degrees), and the memory's outlet ends over that wall, 26 mm left of the throat: its ball drops onto the wall and rolls down it into the throat, beside the carry's path, not across it (`tests/junction.lua`: the carry and the memory's balls in turn, nine of nine right).

- **Under the lane's end.** With the outlet ending 19.7 mm above the throat, the memory's ball stuck at once: rolling down the long wall, it had to pass under the end of the outlet's own right-hand wall, which left 8 mm. It ends 28 mm above the throat now. The MQ's done balls join the same way (a long right wall under MQ2's and MQ3's catches, mirrored), and stuck there the same way, first.
- **COUNT** at first ended over the right half of A1's funnel; its ball stuck under its own outlet's left wall, as the memory's had. Centred over the throat, it worked, but with the pins' friction at 2 or 3 A1 stopped near the middle (+4 to +6 degrees) after the first ball, and the next went the wrong way: the ball has run 40 cm down from MULTIPLY, and came into the top of A1's run at 34 cm/s (the toggle test's ball starts there at rest). A1's funnel has a long right wall too now, and COUNT's outlet ends over it as the memory's do over the left: the ball drops onto the wall and rolls in slowly.
- **The clear lane** starts at CLEAR's drop, right of COUNT now, and bends left at once to pass where COUNT's off balls land. Coming down straight to A1's foot, the clear ball was pushed by the narrowing past its line into A1's wedge (at 0), and with the pins' friction at 0.1 its rubbing dragged A1 over to 1. The lane now overshoots to the left and comes back a little to the right, so the ball meets the foot running along the right wall, as at every other foot.

The memory's switches

The same switches as CLEAR and COUNT, but a 0 has to drop the ball before it can reach the next switch's inlet, only 5 cm below: so the "0" outlet ends at once in a hole just under the flap's pin, its walls round it. The first one stood the hole's inner wall where the divider's slope begins: the ball rested on its end, beside the flap. The wall now begins where the divider's slope ends. A ball dropped there rolls down a wall on the lower level (40 degrees) to the **return lane** down the left edge, and on to the collector, away from the clear lane.

The MQ

Three flip-flops down a staircase like the accumulator's, read the other way: a ball that finds one at 1 (apex left, as it is lettered) sets it to 0 and leaves to the right (done); one that finds it at 0 sets it to 1 and goes on to the next (the borrow). So each ball takes one away, the same parts as the accumulator adding one. A borrow out of MQ3, when the MQ was 0, drops through STOP: the hand watches it (as you watch the machine in MANUAL) and turns no more. The done balls of MQ1, MQ2 and MQ3 run together through the catches under MQ2 and MQ3 (each takes the lane from the one before on its long right wall) and on to M4.

On the board

The tour (README.md): COUNT 0 to 2, CLEAR, ADD 37 + 11 (five balls: 48, then STOP), CLEAR, MULTIPLY 13 x 3 (thirteen balls: 39, then STOP), twenty-two balls. After each ball it checks the accumulator, the MQ, the three distributor flip-flops and STOP against what one more ball should make of them, and that the ball reached the collector. The hand puts balls from the collector back into the input when fewer than four are waiting. It passed in every one of these:

- friction 0.1, 0.3, 0.6
- pin friction 0.1, 0.4, 1, 2, 3
- slope 12, 15, 20, 25 degrees
- tilted across 1 and 2 degrees either way
- 1200, 2400, 4800 steps a second
- the hand's torque on the knob 30, 50, 1000 kg cm^2/s^2

(eighteen runs, each the whole tour). Before the two changes to A1 above (COUNT onto a long wall, and the clear lane's bend), pin friction 0.1, 2 and 3 failed there; the others tried then (friction 0.1 and 0.6, 1200 and 4800 steps, the hand's torque 30 and 1000, slope 25, tilted 2 degrees) passed. `tests/junction.lua` passed (nine of nine) at friction 0.1 and 0.6, pin friction 0.1 and 3, and slopes 12 and 25 degrees.

The computing was right the first time the balls could get through: the distributor, the MQ and the memory gave 48 and 39 once the three places above (the two funnels and the memory's hole) let the balls by.

The printed parts

Every STL (`make-stls.sh`: 14 upper and 14 lower tiles now, 212 mm across) is one watertight solid. Put where they are at rest, no two parts overlap but each foot's stem in its rocker's socket (0.3 mm^3, where they are glued), for all thirteen flip-flops and seven flaps.

Showing where the ball will go

Before each ball the exhibit works out, from what the rockers and switches read, the way the ball will take and what it will do, draws the way as a yellow dashed line on the board (360 short dashes, all alike, moved into place) and says it on the sign (THIS BALL). It is the same reckoning the tour checks each ball against (the accumulator, the MQ, the distributor and STOP after every ball), so a tour that passes did what the line and the sign said, ball by ball.

What it costs

About 430 fixed boxes (320 walls, the plate round its holes, the pegs and posts), twelve balls, thirteen rockers, seven flaps and the wheel. The tour, about 215 simulated seconds (each ball's way is shown for 1.5 s before it goes), took about four minutes of computing headless here (twice the rockers and twice the length of 2a's); between balls, and when it is done, the exhibit rests and costs nearly nothing.

Step 2a: the accumulator

Two levels

E.S.R.'s board has two levels, and so has this one: the upper plate (4 mm) on which the balls compute, and under it the lower level, 15.2 mm high (a ball and 2.5 mm), which takes every ball that is done down to the collector, under everything else. It is what makes a seven-bit accumulator fit: step 1's done lanes each needed a strip of board down to the bins, and seven of them would not have fitted beside the staircase.

- **Done holes.** A ball tipped out of a rocker's right cup drops through a 14.5 mm hole 31 mm right of and 13 mm below the pin, a wall beyond it and below it so it can't run past. With the hole 1.5 mm further out, the flip-flop's test failed at friction 0.6 and at pin friction 3.
- **The collector's wall** is past the end of the plate, with nothing over it. At 15.2 mm high, a ball that had rolled the length of the lower level at a slope of 22 degrees or more jumped it and left the board. It is 30 mm high.
- **The collector** first sloped down to the left: the balls lined up from the left end, over the end of the clear lane, and the ninth ball blocked it, so the tenth stopped in the lane. It slopes down to the right now (2 cm across the board), away from the clear lane and the overflow, and the clear lane ends 35 mm short of its wall, so a ball coming out of it can turn and roll along the wall (ending 10 mm short, two balls were trapped between the lane's end and the wall).

- **The input channel** holds ten balls (13.5 cm); a ball placed past its end fell off the board.

The flip-flop's foot: CLEAR

The original clears its accumulator with a ball on the lower level. Here each rocker has a **foot** on the lower level, joined to it by a stem through a square hole in the plate: an arm 20 mm up the slope from the pin (down the slope, its weight would have made the rocker hang like a pendulum instead of falling over to one side: tried first, it did), ending in a wedge that leans right when the rocker reads 1. A ball coming down the **clear lane** beside it meets the wedge's sloping face and pushes it left, which is the way that turns the rocker to 0. With the foot, the whole flip-flop weighs 5.45 g (rocker 2.5 g, foot 1.9 g, steel ball 1.0 g), its centre of mass 4.4 mm up the slope from the pin: the foot up the slope makes its two leanings deeper than step 1's rocker's (2.4 mm).

- **Pushed part way.** At first the lane was straight, and the ball pushed the wedge only until it could squeeze past it. That left the rocker a little past the middle, where its weight hardly turns it and the pin's friction holds it: at friction 0.6 it stopped at +3 degrees, still reading 1, four times in eleven. And the lane had to be placed within a millimetre either way (13.5 to 15.5 mm right of the pin both failed).
- **The narrowing.** Now the lane's right wall comes in beside the foot, so the ball passes the wedge with its left side 0.5 mm clear of where the wedge is when the rocker is at 0 against its stop. A ball that meets a wedge at 1 must push it all the way back to 0 (the rocker then reads -11 degrees); one that finds it at 0 touches nothing. It works with that gap anywhere from 0.2 to 1.5 mm, at friction 0.1 to 0.6 and pin friction 0 to 4 (`tests/toggle.lua`, eleven or sixteen balls each, from above and down the clear lane in turn), and at slopes of 8 to 25 degrees.
- **Corners.** Two places jammed a clear ball, both found with `DC_CONTACTS`: the lane's left wall coming back just where the right wall narrowed (leaving 12.4 mm for a 12.7 mm ball), and the narrowing starting above the bend where the lane from the flip-flop before joins it (its corner stood in the lane). The left wall now stays open until past the narrowing, and the narrowing starts below the bend. A lane's walls are mitred at its bends.

The switches

A flap 20 mm long and 3 mm thick on a pin at the tip of the divider between two outlets, its free end against one wall of the inlet. It was right ten times in ten (set by hand left and right in turn, then a ball) at friction 0.1 and 0.6, pin friction 0 and 2, slopes 8 and 25 degrees, and with the ball run in from 1 or 15 cm above (`tests/switch.lua`). The divider's tip must stay 5.5 mm below the flap's pin: at 1.5 mm the flap jammed against it.

AUTO: the START lever (the escapement works; not on the board yet)

The original's AUTO lets a ball reaching the bottom release the next from the top, through the START lever. A long lever on the lower level was tried, its top end carrying two teeth up through slots into the input channel (an escapement: one ball out each time the lever is pushed) and its bottom end a wedge in a trip lane. It failed: round teeth coming in from the sides of the channel are pushed sideways by the balls they hold, and with the lever's pin 9 cm away that was enough to hold the lever pushed (it never came back after the hand let go), and a trip ball let all six balls out at once. The next design needs teeth whose faces meet the balls square on, on a lever whose pin is in line with the channel, so the balls' weight turns it neither way.

Second attempt (2026-10-27): the wheel held by an anchor. The feeder wheel itself as the escapement: let the balls' weight turn it, and hold it a quarter turn at a time with a clock's anchor (two fingers on a pin to the wheel's right, catching four pins on an escape star above the balls). Each finger lay along the line its pin moves on, through the anchor's pin, so the wheel's push could not turn the anchor.

In a test of the wheel and anchor alone, the anchor (a 1/2 inch steel ball its counterweight) swung to tripped when pushed and back by itself when let go (not quite home: against a pin that had stopped short). But the wheel would not turn:

- **Balls coming down onto the top of the wheel** (as now): the next ball in the channel dips into the notch as the ball in it is carried away, and the notch's edge must then push it, and every ball above it, back up the channel. The hand's knob can (it is allowed 50 kg cm^2/s^2); the weight of one ball in the wheel gives about 3, and the wheel stopped 12 degrees round.
- **A paddle wheel fed from the side,** the channel coming down in line with its left-hand space so the balls' weight pulled it round: the next ball still follows the one being carried away, and the vane coming round next must sweep through the channel and push it back. It stopped 58 degrees round, a ball jammed between the vane and the channel's wall.

Whatever lets one ball go must at some point push the next back against the balls above it. So that push has to come from somewhere other than the balls in the wheel: from the trip (a ball arriving at the bottom has fallen over a metre, but loses most of that at the collector's wall), or, more surely, from a weight the hand winds up, as a clock's, turning the wheel through the anchor, which already worked. That is the next thing to try.

Third attempt (2026-10-27): a wound weight, a fly, and straight fingers. The wheel driven by a steady torque (the weight) with a top speed (a fly, the air brake on a clock's striking train), and held by the anchor (digicomp2-geometry.lua, G.ANCHOR; tests/escapement.lua). It works: at rest nothing comes out, and each trip, from a tenth of a second to two seconds long, lets out exactly one ball and the anchor comes back home. What it took:

- **Each finger on its own side of the pin it stops.** The first finger stops a pin coming toward the anchor's pin, and lies on the near side of the stopping point; the second stops a pin moving away, and must lie beyond it, on a bar carried over the pins. With both on the near side the second never touched its pin.
- **Straight fingers, not curved ones.** Fingers curved as a deadbeat clock's (arcs about the anchor's pin, so the pin slides along them while the anchor swings) held the anchor tripped whenever the wheel was driven: the pin caught on the arc's edge, and the push there no longer passed through the anchor's pin (it turned the anchor half as hard as the counterweight, the other way). Straight fingers along the line through the anchor's pin have no such edge.
- **Rollers on the fingers** (no friction at the anchor): it passed with friction 0, 0.05 and 0.1 there; with the curved fingers any friction held it.
- **A weight big enough to push the column back.** At the board's 15 degrees, a torque of 20 to 60 kg cm^2/s^2 on the wheel works and 10 does not (the wheel can't push the next ball back up the channel); at 25 degrees 30 is too little for a short trip, and 50 works.
- **A fly.** With the top speed 1.5 to 12 radians a second (a quarter turn in a fifth of a second to a second) every trip let out one ball; with no limit one trip in eight let none.

It passed (eight trips each) with friction 0.1, 0.3, 0.5; pin friction 0.1, 0.4, 2; slope 8, 15, 20 degrees; 4 and 9 balls in the channel; the hand's push 100 and 300.

Still to do before AUTO is on the board: the weight, its winding and the fly as parts (here a torque and a speed limit on the wheel's pin); the trip, a ball arriving at the bottom pushing the anchor's arm across; STOP, whose ball must not; and the AUTO/MANUAL switch between the hand's knob and the weight.

The accumulator on the board

The tour (README.md): COUNT from 0 to 4; 125 to 127 and over to 0, the carry running through all seven; CLEAR from 90 and from 127; CLEAR at 0; both switches off. Every reading and every ball's arrival in the collector checked, eleven balls. It passed in every one of these:

- friction 0.1, 0.2, 0.3, 0.4, 0.5, 0.6
- pin friction 0.1, 0.4, 1, 2, 3 (a quarter of the estimate to seven times it)
- slope 8, 10, 15, 20, 25 degrees (at 8 the tour takes longer)
- tilted across 1 and 2 degrees either way
- 1200, 2400, 4800 steps a second
- the hand's torque on the knob 30, 50, 1000 kg cm^2/s^2

and failed with no pin friction at all (the 1s rocker bounced back off its stop, on the board though not in its test; any real pin has some), and tilted 3 degrees across either way (as in step 1, the balls are carried past the apexes). So: level across within 2 degrees.

The printed parts

Every STL (`make-stls.sh`) is one watertight solid. Put where they are at rest (`PART="rest-..."` in `parts/digicomp2.scad`), no two parts overlap but each foot's stem in its rocker's socket (0.3 mm^3, where they are glued). The bottom lower tiles first came out with the collector's wall cut off at the plate's height, and the collector's tall end walls overlapped the plate's rim; both fixed.

What it costs

A few hundred fixed boxes (walls, the plate round its holes, pegs, posts), twelve balls, seven rockers, two flaps and the wheel. The tour, about 90 simulated seconds, took about a minute of computing headless here; the exhibit rests (stops stepping) whenever nothing has moved for half a second, so between balls it costs nearly nothing.

Step 1: the counter (2026-10-26)

Three flip-flops on a one-level board, with lanes to bins for the done balls. The rocker then had no foot, and `tests/toggle.lua` was the rocker alone under its channel (it is now the rocker with its foot: see step 2a).

1/2 inch steel balls (12.7 mm, 8.35 g), PLA parts, friction 0.3, no bounce, the board at 15 degrees, 2400 steps a second, the pins' friction 0.4 kg cm^2/s^2, unless a line says otherwise. The flip-flop was worked out alone first, with `tests/toggle.lua` (one rocker under its channel, balls let go one at a time, 16 of them), then in the counter, with its tour (eight balls from 0, each reading checked).

The flip-flop

The idea. A rocker on a pin whose centre of mass is up the slope from the pin is over-centre: it falls to one side or the other against a stop and stays, a bit of memory. Its top is a roof whose apex is up the slope from the pin, so the apex leans with it. A ball coming straight down meets the roof beside the apex, on the side that is raised, and its weight on that side takes the rocker over. So each ball goes the other way from the last. The first rocker, an inverted V of plain strips, did this sixteen times out of sixteen at every slope from 6 to 30 degrees and every friction from 0.1 to 0.6, as long as it leaned at least 8 degrees (at 6 degrees the apex moved too little to choose a side for the ball).

Cups. With friction at its pin, the plain roof stopped half way over: the ball runs off the end of the roof while the rocker is still near the middle, where its own weight hardly turns it, and the pin's friction holds it there. The next ball then finds no raised side. At 0.8 kg cm^2/s^2 of pin friction, 14 of 16 balls went the wrong way. A **lip** at each end of the roof makes a cup, which holds the ball until the rocker has gone over; at the end of the swing the cup tips it out. With 5 mm lips the rocker worked at four times that friction. The lips lean out 67 degrees from straight up the slope:

Lips (lean out)	Counter's tour, under 10 hard conditions
none	fails at 0.8 kg cm^2/s^2 of pin friction (the rocker alone)
60 degrees	8 of 10: a ball stays in a cup when the table is tilted 2 degrees across
67 degrees	9 of 10: with no pin friction at all, a rocker bounces back from its stop
75 degrees	9 of 10: the same
3.5 mm long, 67 degrees	8 of 10: too short to hold the ball over at high pin friction

(The ten: tilted 1 and 2 degrees across either way, pin friction 0 and 2, friction 0.1 and 0.6, slope 10 and 20 degrees. These were run with the pin friction first estimated, 0.25 kg cm^2/s^2, before the steel ball below, which these rockers did not have.)

The weight. The bouncing back is a sign that the rocker's two leanings are too shallow: its centre of mass was only 1.5 mm up the slope from the pin, so the dip it rests in is tiny, and a rocker coming hard onto its stop can rebound right out of it. A **steel ball of 1/4 inch (1.04 g)** pressed into the rocker under its apex, 5.8 mm up from the pin, moves the centre of mass to 2.4 mm and makes the dip deep enough: with it and the 67 degree lips, the counter passed all ten (and so did 1 and 2 g, 5 mm up, tried before it).

The apex. The first rockers were built of boxes, and the strut from the pin to the apex stood 3.5 mm above the roof: the ball landed on its corner and went either way. The rocker is now built of convex pieces whose upper edges are the roof's lines exactly, nothing above them (and the printed part is the same pieces).

Its numbers. Roof 32 mm wide at 30 degrees, apex 10 mm up the slope from the pin, lips 5 mm, strips 3.5 mm, 9 mm high, 1.5 mm off the board on the pin's boss; it leans 12 degrees either way to its stop pegs. It weighs 2.5 g printed solid, 3.6 g with its steel ball.

The pin's friction. The rocker lies on the boss round its pin with its weight across the board, 3.4 g-force; PLA on PLA (0.3 to 0.4) at the boss's mean radius (about 2.9 mm) gives about 0.4 kg cm^2/s^2 (0.04 mN m), the pin itself a tenth of that. The simulation uses 0.4. At that, the rocker's own weight cannot turn it against the pin when it is near its stop (about 0.03 at 8 degrees), so it stays where the ball leaves it, 8 to 12 degrees over: that is still far enough for the next ball to choose its side.

The rocker alone (`tests/toggle.lua`, 16 balls each; the least lean any ball left it at):

	Passed	Least lean
as built	yes	9.1 degrees
pin friction 0, 1, 2, 4, 5	yes	11.6, 11.0, 10.3, 3.8, 0.5
pin friction 6	**no**: the rocker stops in the middle	1.3
friction 0.1 to 0.8	yes	7.8 to 9.8
slope 6 to 30 degrees	yes	9.6 to 11.6
run 5, 10, 15 cm longer (faster balls)	yes	7.6, 6.0, 2.8
run 20 cm longer	**no**: the ball comes too fast and jumps the cup	
let go up to 0.75 mm off centre (all the channel allows)	yes	8.7

	Passed	Least lean
tilted across 2 or 3 degrees one way, 2 the other	yes	7.1 to 10.0
tilted across 3 degrees the other way, or 4	**no**	
1200 and 4800 steps a second	yes	8.8, 9.9

Tilted across, the ball runs down its channel pressed against the downhill wall, and the board's sideways pull carries it past the apex onto the low side, which it then rolls down without turning the rocker. So the board must be level across: within 2 degrees.

The counter

The staircase. Each flip-flop 3.5 cm left of and 8 cm below the last, so a carry leaving the left cup falls straight into the next one's funnel; each done ball leaves the right cup into a lane straight down to the bins. The lanes' mouths were first 2.4 cm right of the pin: a ball came out of the cup and stuck, wedged between the rocker's lip and the lane's flared wall (found with `DC_CONTACTS`). At 2.8 cm they clear.

The feeder. A wheel of four notches turning under the channel of balls. Three things were found:

- A wall's box runs 1.5 mm past its ends (so walls meet solidly), and the ends of the channels at the wheel's housing reached into the wheel's path: the wheel rubbed them and stuck at friction 0.5. Those channels now start 2 mm further out.
- The hand must be able to push: seven waiting balls press on the wheel's rim. The simulation's hand turns the knob with up to 50 kg cm^2/s^2 (5 mN m, a light finger); it also passed with 20 and 1000.
- In the first wheel the teeth only touched the hub, which would have printed as five pieces: the hub is now 0.5 mm wider than the notches, so the wheel is one solid.

The tour, eight balls from 0, each reading and each ball's bin checked, the machine as built (pin friction 0.4), passed in every one of these:

- friction 0.1, 0.2, 0.3, 0.4, 0.5, 0.6
- pin friction 0, 1, 2, 3 (eight times the estimate)
- slope 8, 10, 15, 20, 25 degrees
- 1200, 2400, 4800 steps a second
- tilted across 1 and 2 degrees either way
- friction 0.6 with: tilted 2 degrees either way, slope 10 degrees, pin friction 2; friction 0.1 with: tilted 1 degree either way or 2 one way, slope 10 degrees, no pin friction
- the hand's torque 20 and 1000

and failed at pin friction 4 (a rocker stopped near the middle), tilted 3 degrees across either way, and at friction 0.1 tilted 2 degrees the other way. So: level across within 2 degrees, and within 1 to be sure.

What it costs

Little: eight balls, three rockers and the wheel, against some fifty fixed walls and pegs. The tour, about 28 simulated seconds, took 7 s of computing headless here; the exhibit rests (stops stepping) once nothing has moved for half a second.

What is idealised

- The knob is turned by the hand: a motor on the wheel's pin, exactly a quarter turn at a time. In the machine as built you turn it by its four marks (how far off a quarter it may be was not tried).
- The pins are hinges with a constant friction torque.

- Setting it back puts the balls and the rockers in place by hand.